



EAST TEXAS A&M
— UNIVERSITY —

CSCI 380.61E WEB PROGRAMMING AND INTERFACE

OVERVIEW

- **Web Services**
- What do we mean by Web Services?
- What is **SOAP** and **REST**?
- Limitations and comparison
- Questions and discussion

REVISION QUESTION

Fix the bug

```
function getMessageText() {  
    return $( '#message' ).innerHTML();  
}  
  
document.body.text = '<div id="message">The message</div>';  
console.log(getMessageText());
```

ANSWER

innerHTML and **text** (used incorrectly)

```
function getMessageText() {  
    return $( '#message' ).text();  
}  
  
document.body.innerHTML = '<div id="message">The message</div>';  
console.log(getMessageText());
```

REVISION QUESTION

Which of the following is AngularJS:

- a) Library
- b) Toolkit
- c) Framework
- d) Operating System

ANSWER

c) Framework

REVISION QUESTION

Which of the following is jQuery:

- a) Library
- b) Toolkit
- c) Framework
- d) Operating System

ANSWER

a) Library

REVISION QUESTION

What does MVC stand for?

- a) Model View Controller
- b) Message Verification Certificate
- c) Model Vender Certified
- d) Main View Control

ANSWER

a) Model View Controller

Model-View-Controller (MVC) is a design pattern for separating an application into three main logical components

WHAT ARE WEB SERVICES?

WHAT ARE WEB SERVICES?

- Web services are **applications** that allow for **communication** over the internet
- There are a number of **protocols** for web services to communicate
 - Simple Object Access Protocol(**SOAP**)
 - Representational State Transfer (**REST**)

WHY ARE WEB SERVICES **IMPORTANT**?

Web Services **enable** applications to **talk** to one another

Low Cost (just use Internet)

Reusable/ Open Data

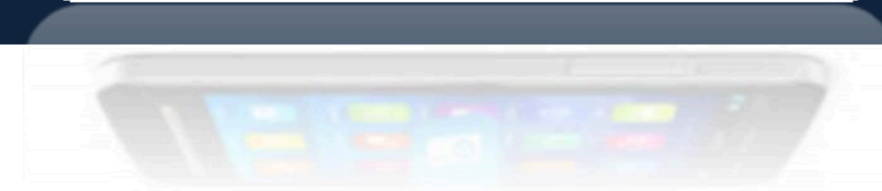
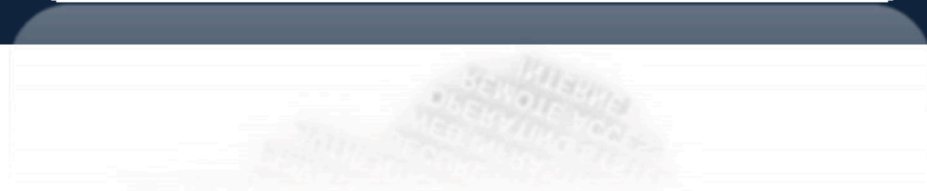
Standard communication protocols

Open and ready to talk

Available to all sizes of organisations and individuals



Web Services “is” a fundamental technology of Cloud Computing and Mobile Computing plus Robotics, Augmented Reality, etc.



WEB SERVICES

- Allows **different** applications to communicate
- Standard **data-exchange format**
- Many Web Applications provide a web service access to their resources
- Can enrich web apps by resources offered by web services

WEB SERVICES

- Many types of applications and users
- Many types of devices
- Challenges of data syncing
- More users == more business

DO WEB SERVICES PROVIDE **USER INTERFACES**?

DO WEB SERVICES PROVIDE USER INTERFACES?

NO

Web services **do not** provide UIs as they are not meant to be used by human end users

WHAT WEB SERVICES ARE AVAILABLE?

WEB SERVICES SOLUTIONS

1. A non-standard hacky way and implications

2. **SOAP**

- Simple Object Access Protocol

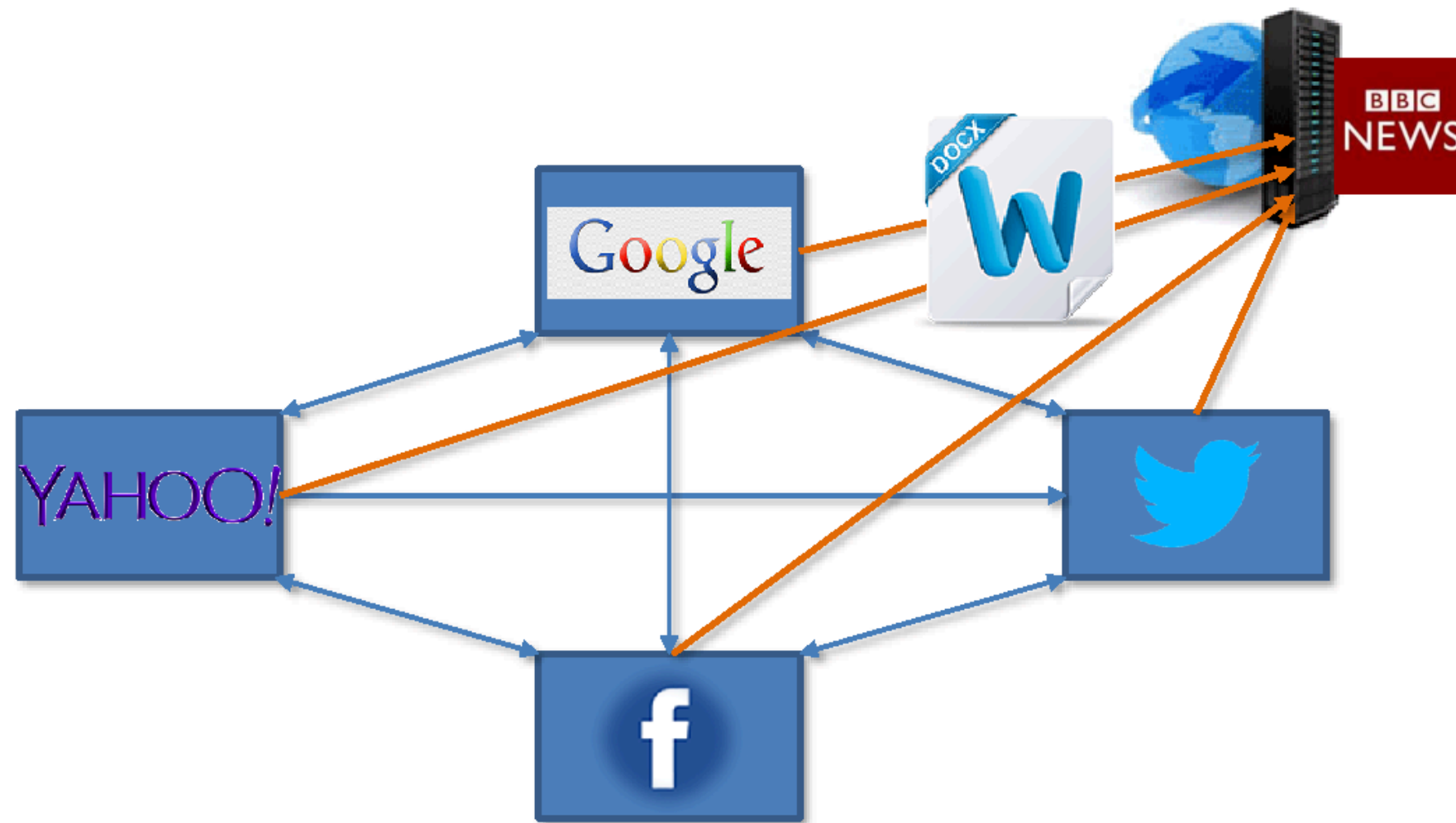
3. **REST**

- REpresentational State Transfer

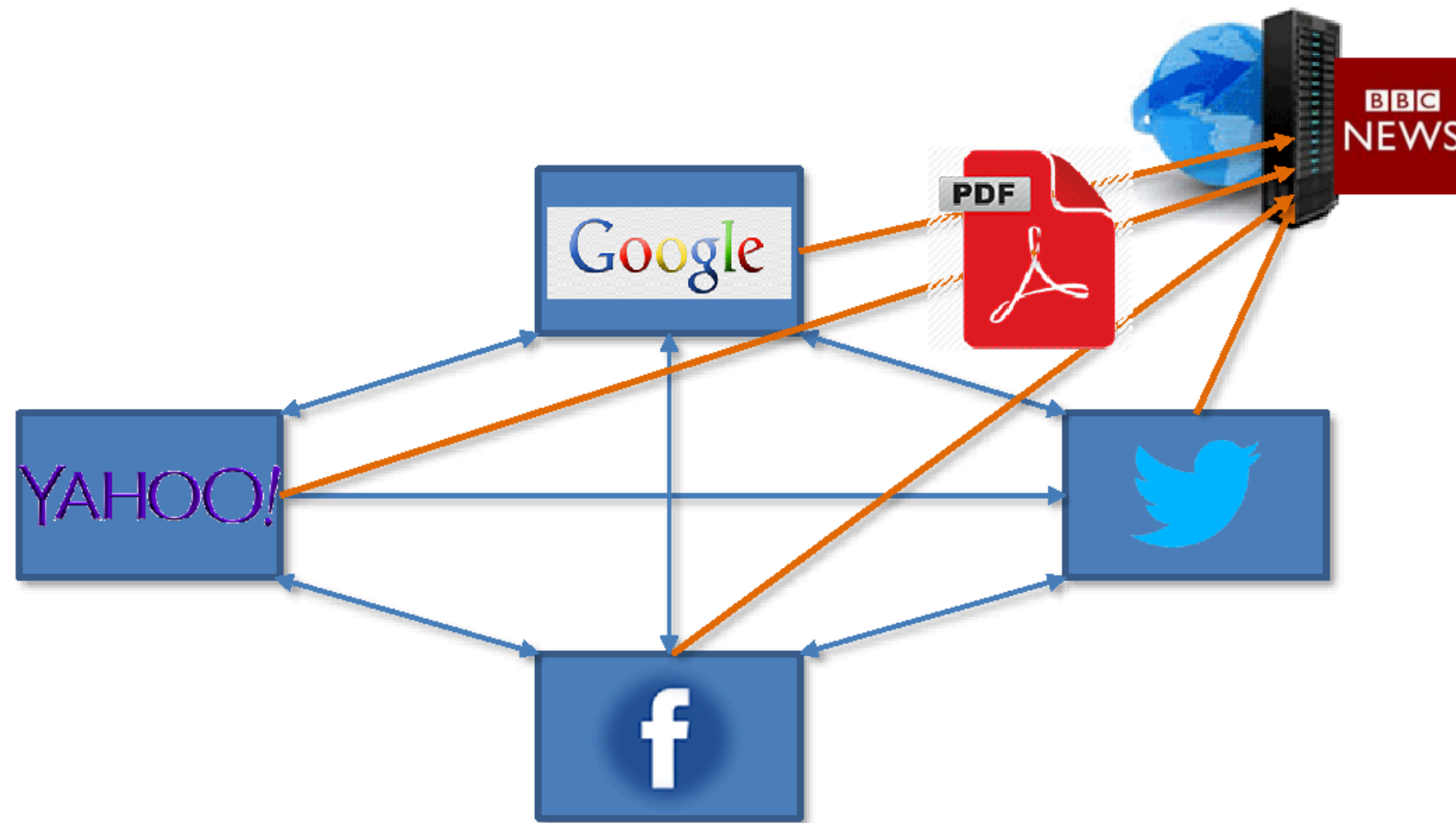
WHEN DO WEB SERVICES BECOME IMPORTANT?

(data exchange increases in complexity)

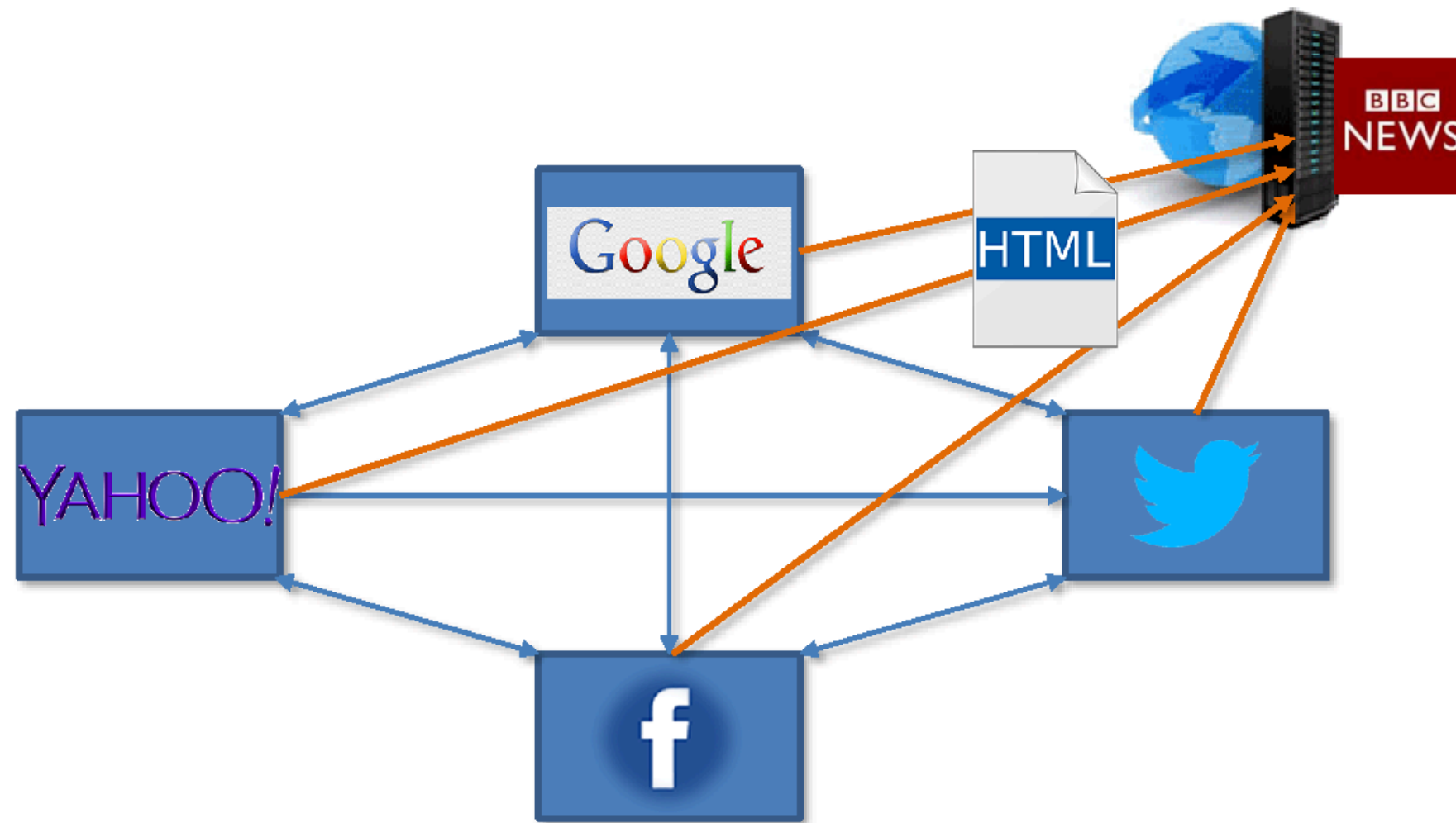
Web Services need to exchange data



Web Services need to exchange data

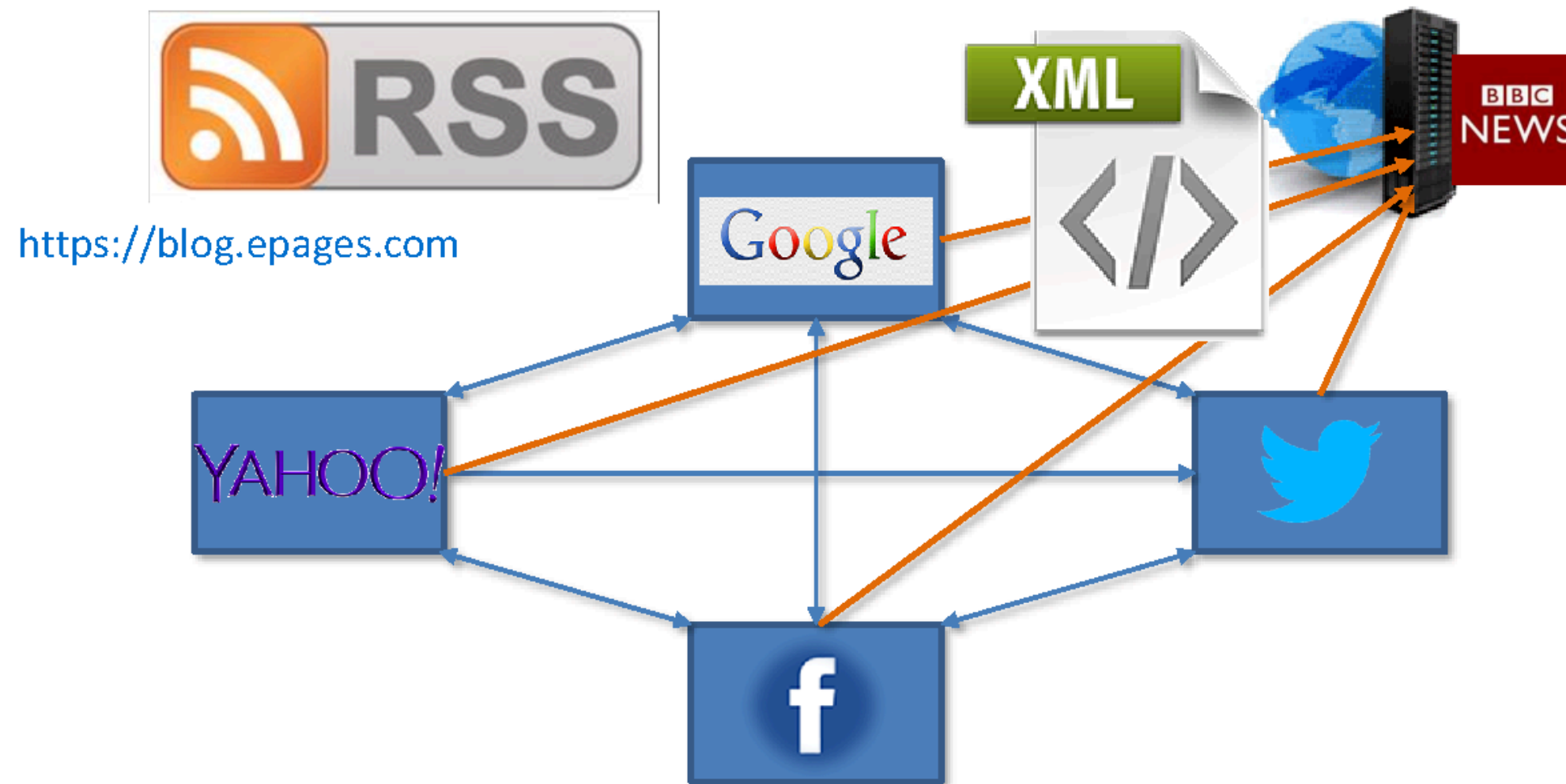


Web Services need to exchange data



Web Services need to exchange data

- We need a common standard data-exchange format



WHAT IS SOAP?

SOAP

- SOAP stands for **Simple Object Access Protocol**
- SOAP is a **standard** for web-services **access protocol**
- SOAP messages are encoded in a common XML format
- Can use different protocols to transport a SOAP message
 - HTTP, FTP, BEEP, ...

SOAP : HOW IT WORKS

SOAP DETAILS

- SOAP is a **protocol**
- Only support **XML** format
- Versatile
- **Powerful/Complex**
- Automated tools exist
- Often seen as **Verbose**

WHAT IS REST?

REpresentational **S**tate **T**ransfer

REST

- Is an **architectural approach** for distributed systems
- Have 6 guiding constraints including:
 - **Client-Server, Stateless, Uniform Interface, ...**
- Information is abstracted as **resource** identified by a URI (Uniform Resource Identifiers)
 - document, image, service, collection of other resources,...
- Different **representations** for a resource are possible: XML, JSON, PDF, JPEG, ...
- Another concept is **resource methods** to perform transition
 - a popular choice is HTTP GET/PUT/POST/DELETE

REST : HOW IT WORKS

REST DETAILS

- Uses **HTTP** methods
- Directory like structure
- Uses proper MIME types
- Human readable result
- **Lightweight**
- **Easy** to build (no toolkits required)

WHAT IS A **REST**FUL WEB SERVICE?

WHAT IS A **RESTFUL** WEB SERVICE?

- RESTful web service (i.e., web service that is created and accessed using **REST principles**)
- use **HTTP** protocol methods for operations
- For example, **HTTP methods** (post,get,put,delete) to manage **operations**

REST Web Services

HTTP methods

GET : retrieves data from the service

POST: submits data to the service (to retrieve data too)

PUT: creates or updates data (resource) on the service

DELETE: deletes data (resource) from the service

OPTIONS: requests information

TRACE: debugs the m2m communication

HEAD: a GET response without body

CONNECT: tunnels connection to a proxy

PATCH: partially updates data on the service

REST Web Services

URI: a URL with style

In REST, URLs identify resources (URIs)

<http://www.boeing.com/aircraft/747>

~~<http://www.boeing.com/index.php?what=aircraft&model=747>~~

~~<http://www.boeing.com/aircraft/747/get>~~

1 **GET** /aircraft/747 HTTP/1.1

request

2 Host: boeing.com

1 HTTP/1.1 200 OK

response

2 Content-Location: index.php

3 Content-Type: application/json

4 Content-Length: 1234

5

6 {"aircraft": "747", "specs": {"engine": "R&R B1",
"speed": "555mph", "seats": 366 } }

```
1 PUT /aircrafts HTTP/1.1
2 Host: boeing.com
3 Authorization: UID {auth token}
4 Content-Type: application/json
5 Content-Length: 1234
6
7 { "aircraft": "727", "specs": {"engine":P&D 7,
  "speed":400mph, "seats":280} }
```

request

```
1 HTTP/1.1 201 OK
2 Content-Location: index.php
3 Content-Type: application/json
4 Content-Length: 123
5
6 { OK }
```

response

KEY PRINCIPLES OF REST

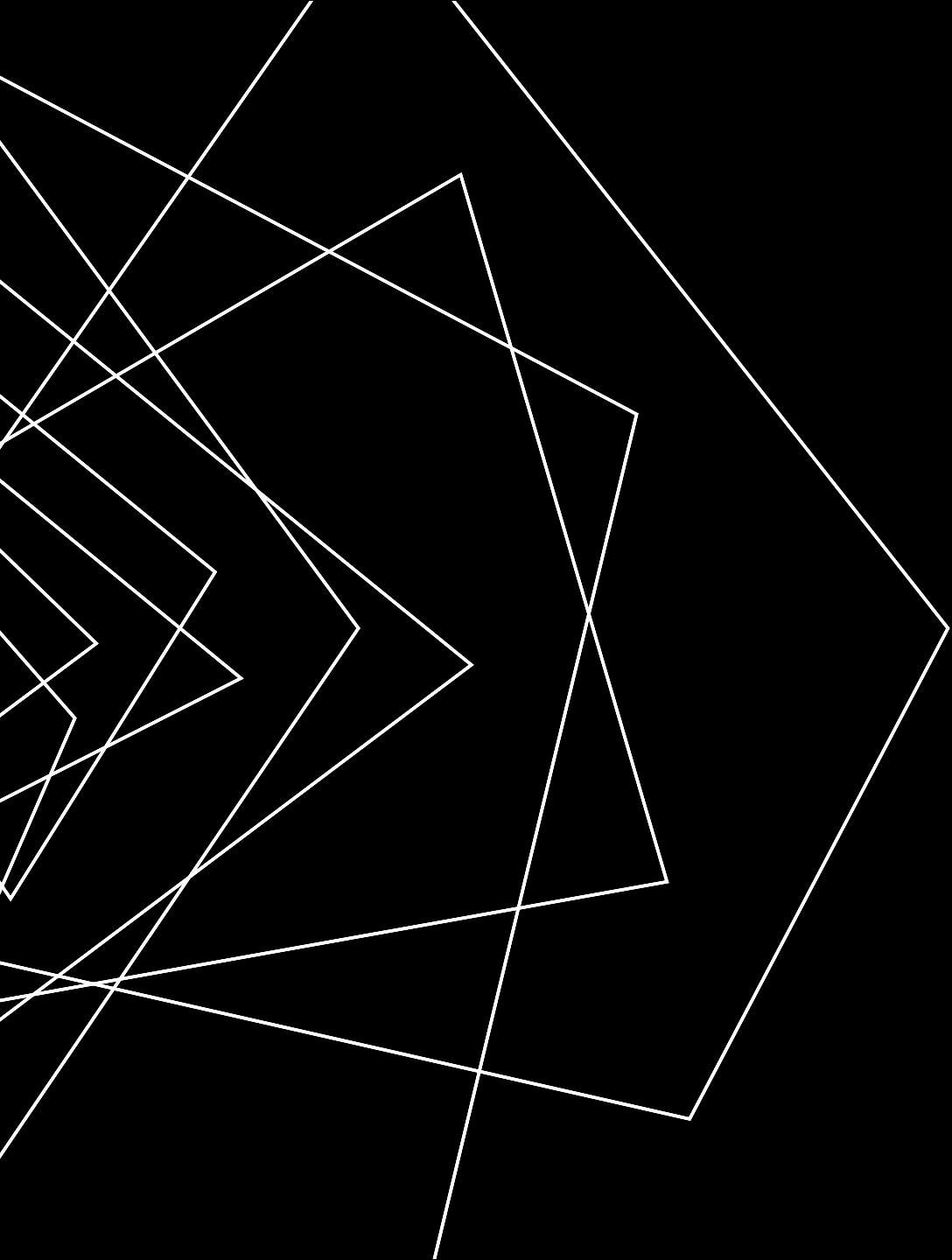
- Everything is a **resource**
- Every resource is identified by a **unique identifier**
- Uses a simple and uniform interface
- **Stateless**

REST OR SOAP WEB SERVICE

- REST exposes **resources** that represent data, while SOAP exposes **operations** that represent logic
- REST emphasises **simple p2p** communication, while SOAP emphasises **loosely coupled** distributed message
- REST supports **multiple data formats**, while SOAP encodes everything in XML with data attachments
- REST emphasises **stateless communication** while SOAP supports stateless and stateful operations
- REST is **lightweight** compared to SOAP
- REST depends on other security approaches (e.g., OAuth)

SUMMARY

- Understand basic Web Service **concepts**
- Why web services are important
 - **Communication**
- Whats and Whys of **SOAP** and **REST**



THANK YOU

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